

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B.Tech. Examination (CE/IT)

November 2013

IT 201 Database Management System

Date: 28.11.2013, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Draw and explain the structure of Database System. [05]
(b) Justify: "Data redundancy and inconsistency" is the disadvantage of file system over database system. [02]

- Q - 2 (a) Construct an E-R diagram for university registrar's office which maintains following data. [05]

- 1) course including number, credit, syllabus
- 2) course offering including course number, year, sem, instructor
- 3) students including std_id, name, program
- 4) instructor including ins_id, name, dept, title

- (b) What is the use of "Aggregation" in E-R model? Explain with suitable example. [05]
(c) Which are the different types of query languages in database system? Explain each in detail. [04]

OR

- (b) What is weak entity set in E-R model? Explain with suitable example. [05]
(c) What is the need of 'Key' in database system? Differentiate Super key, Candidate Key and Primary key with example. [04]

- Q - 3 (a) Suppose you are given a relation $R = (A, B, C, D, E)$ with the following functional dependencies: $F = \{CE \rightarrow D, D \rightarrow B, C \rightarrow A\}$. [06]

- a. Find key attributes from above relation.
- b. Check whether R is in 2NF? If no, covert R into 2NF.

- (b) Give a relational algebra expression for below queries, [05]

Item (ino, description, unit price)

Supplier (sno, sname, address)

Supplied (sno, ino, sdate, quantity, per_unit_discount)

- 1) Find the item number having unit price greater than 1000.
- 2) List the supplier number and name along with supplied quantity.
- 3) Rename the relation item with new_item.
- 4) Display all the information about item which is supplied by the supplier.
- 5) List the total quantity for EACH item supplied.

- (c) Find the canonical-cover / irreducible-set for following FD. [03]
Relation R = (A, B, C, D)
 $F = \{ A \rightarrow BC, B \rightarrow C, AB \rightarrow D \}$

OR

- (c) For Relation $R = \{A, B, C\}$, where $F = \{A \rightarrow B, B \rightarrow C\}$. Check whether following decompositions are lossless join decomposition or not. [03]
- 1) $R_1 = (A, B), R_2 = (B, C)$

SECTION – II

- | | | |
|-----------|--|------|
| Q - 4 (a) | Explain the basic steps for query processing in detail. | [05] |
| (b) | Define the shared-mode lock and exclusive-mode lock in concurrency control. | [02] |
| Q - 5 (a) | Draw and explain the state diagram of transaction in detail. | [04] |
| (b) | Write down the four cases of conflict serializability for swapping of two consecutive instructions. | [04] |
| (c) | Explain growing phase and shrinking phase of the two-phase locking protocol in detail with suitable example. | [06] |

OR

- (c) What is deadlock in concurrency control? Explain techniques to prevent the deadlock. [06]
Q - 6 (a) Consider the below schedule, [04]

T1	T2
R(A)	
W(A)	
	R(A)
	W(A)
R(B)	
W(B)	
	R(B)
	W(B)

- 1) Using precedence graph, check whether above schedule is conflict serializable or not.
 - 2) If above schedule is conflict serializable then make equivalent serial schedule for that.
- (b) Define the following terms of database system: [04]
- 1) Starvation
 - 2) Domain
 - 3) View
 - 4) Recoverable schedules
- (c) Explain the types of errors that cause transaction failure. Explain the recovery system of database. [06]

OR

- (c) Explain discretionary access control and mandatory access control in database security. [06]
